

A Multi-Source Satellite Imagery Framework for Water Body Segmentation and River Discharge Estimation

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Abstract—Continuous quantification of open water resources and fluvial discharge dynamics is indispensable for mitigating catastrophic flood inundation, managing regional water allocations, and preserving riverine ecology. Traditional hydrometric regimes rely heavily on physical stream gauges; however, in-situ networks are sparsely distributed across remote basins, demand extensive capital maintenance, and frequently suffer physical washout during peak flood stages. Although orbital Earth observation offers synoptic surveillance, standard operational methodologies remain constrained by cloud attenuation in optical swaths, radar speckle artifacts, mixed-pixel degradation along narrow channels (<30 m), and an inability to detect floodwaters concealed beneath dense tree canopies or predict streamflow across ungauged catchments.

To resolve these limitations, this study presents an end-to-end framework fusing multi-mission optical (Sentinel-2 MSI, PlanetScope 3 m, Landsat Next) and polarimetric Synthetic Aperture Radar (SAR; Sentinel-1 C-band, NISAR L/S-band) imagery with physics-guided machine learning. An Attention U-Net model with spatial gating filters is developed to segment complex river boundaries under persistent overcast, effectively suppressing cloud shadows and specular terrain noise. Morphological skeletonization extracts the river centerline and casts orthogonal cross-sectional transects to determine dynamic reach widths (W), while dual-polarization Radar Vegetation Index (RVI) decomposition isolates dihedral double-bounce reflections to map sub-canopy inundation. For volumetric flow estimation, we formulate an uncertainty-weighted hybrid model coupling Manning’s open-channel hydraulic formulation with ensemble machine learning regressors, calibrated with 95% confidence intervals. An ungauged basin solver leverages Leopold–Maddock regional hydraulic geometry scaling, and real-time hydrometeorological telemetry (precipitation, soil moisture, reservoir releases) is incorporated for 36-hour predictive flood warning. Validated across nine operational stations in the Ganga, Brahmaputra, Godavari, Krishna, Mahanadi, and Kosi basins, the system achieves an Intersection over Union (IoU) of 92.4%, a Dice coefficient of 0.958, and a discharge Nash–Sutcliffe Efficiency (NSE) of 0.932, while maintaining a lightweight CPU runtime of <95 ms for resource-constrained agencies.

Index Terms—Attention U-Net, Synthetic Aperture Radar (SAR), River Discharge Estimation, Manning’s Hydraulics, Physics-Informed Machine Learning, Multi-Source Fusion, Inundation Mapping, Ungauged Basins, Early Warning Systems.

I. INTRODUCTION

Fluvial discharge (Q) represents the fundamental mass transfer flux governing continental hydrology, sediment routing, and ecosystem stability. Driven by intensifying climate change, land-use transitions, and hydrometeorological

extremes, river systems worldwide are experiencing severe seasonal inundations, rapid channel avulsions, and structural embankment failures [1], [2]. Establishing resilient, high-frequency streamflow surveillance is vital for emergency disaster response, municipal planning, and reservoir barrage scheduling.

In-situ stream gauging infrastructures (e.g., stage rating stations and acoustic Doppler current profilers) have historically served as the benchmark for hydrometric measurement [2]. Nevertheless, terrestrial gauge networks present acute structural drawbacks: spatial distributions across transboundary and mountainous headwaters remain sparse, cross-border gauge telemetry is often restricted, and physical gauge sensors are frequently submerged, damaged, or swept away during catastrophic flood surges [5]. These vulnerabilities create critical telemetry blackout periods during the most hazardous phases of extreme flood events.

Orbital remote sensing offers an indispensable, non-invasive methodology capable of delivering repetitive synoptic coverage over extensive drainage basins. Multispectral optical sensors, such as the Sentinel-2 Multispectral Instrument (MSI), provide spectral band combinations optimized for delineating open water surfaces via normalized band ratioing (e.g., NDWI, MNDWI, AWEI) [3], [4]. Nevertheless, optical observations are fundamentally incapacitated by dense monsoon cloud cover and atmospheric haze, precisely when flood monitoring is most urgent. Spaceborne Synthetic Aperture Radar (SAR) systems, such as Sentinel-1 C-band and NISAR L/S-band polarimetry, circumvent atmospheric interference by emitting microwave pulses that penetrate clouds and rain day and night [5]. Because smooth water bodies act as specular reflectors, backscatter intensity is scattered away from the sensor, appearing as distinctive dark features ($\sigma^{\circ}_{VV} < -18$ dB). However, pure SAR interpretation remains plagued by multiplicative speckle, wind-induced capillary wave roughening (which elevates backscatter and mimics land), and severe layover/shadow artifacts in undulating terrain [6].

Beyond sensor-level trade-offs, three major hydrological bottlenecks persist in spaceborne river monitoring:

1) *Narrow Channels and Braided Tributaries*: River channels with reach widths smaller than standard satellite pixel resolutions (<30 m) suffer from mixed-pixel spectral averaging, leading to severe reach width underestimation [7].

2) *Inundation Beneath Vegetated Canopies*: Floodwaters expanding into riparian woodlands and flooded agricultural plots are obscured from optical sensors by foliage, while single-polarization SAR experiences diffuse volume scattering, masking the underlying flood boundary [8].

3) *Ungauged Basin Generalization*: Purely empirical machine learning regressors require extensive historical gauge calibration records; when applied to unmonitored reaches, data-driven models suffer from catastrophic generalization failure due to unconstrained physical drift [9].

To overcome these challenges, this study presents a fully realized, multi-source satellite framework for water-body segmentation and river discharge estimation. The framework integrates Sentinel-1 SAR and Sentinel-2 multispectral imagery alongside high-resolution PlanetScope (3 m) and NISAR polarimetric observations. An Attention U-Net model performs pixel-level water segmentation with spatial gating, after which geometric features such as water surface area and dynamic river width are extracted. Discharge is quantified using an uncertainty-weighted hybrid model coupling Manning’s hydraulics with machine learning regressors, calibrated with 95% confidence bounds. Regional hydraulic geometry enables discharge prediction in ungauged basins, and real-time telemetry affords 36-hour flood lead-time alerts.

The primary contributions of this work are:

- 1) Formulating an all-weather multi-source ingestion pipeline combining optical (Sentinel-2, PlanetScope, Landsat Next) and SAR (Sentinel-1 C-band, NISAR L/S-band) data.
- 2) Implementing an Attention U-Net architecture with soft spatial gating to segment river boundaries while suppressing cloud shadows and specular terrain noise.
- 3) Extracting dynamic channel reach widths (W) via morphological centerline skeletonization and orthogonal transect profiling, coupled with polarimetric Radar Vegetation Index (RVI) decomposition to resolve sub-canopy inundation.
- 4) Developing an uncertainty-weighted hybrid streamflow model fusing Manning’s hydraulics with ensemble machine learning regressors, equipped with 95% confidence intervals.
- 5) Formulating an ungauged basin solver based on Leopold–Maddock regional hydraulic geometry, alongside real-time rainfall, soil moisture, and reservoir release coupling for 36-hour flood warning.
- 6) Evaluating the system across nine operational stations in six major Indian basins, with a lightweight CPU mode (<95 ms) tailored for resource-constrained disaster agencies.

II. RELATED WORK

A. Spectral Indices for Water Surface Extraction

Delineating open water from multispectral imagery traditionally relies on band arithmetic exploiting the sharp contrast between high reflectance in green wavelengths and near-zero reflectance across near-infrared (NIR) and short-

wave infrared (SWIR) spectra. McFeeters [3] formulated the Normalized Difference Water Index (NDWI):

$$NDWI = \frac{\rho_{Green} - \rho_{NIR}}{\rho_{Green} + \rho_{NIR}} \quad (1)$$

To suppress urban and soil reflectance, Xu [4] proposed the Modified NDWI (MNDWI), replacing NIR with SWIR:

$$MNDWI = \frac{\rho_{Green} - \rho_{SWIR}}{\rho_{Green} + \rho_{SWIR}} \quad (2)$$

Feyisa et al. [10] developed the Automated Water Extraction Index (AWEI) to suppress topographic mountain shadows. While efficient under cloud-free conditions, optical indices fail completely under overcast skies, necessitating radar fusion.

B. Radar Polarimetry and Canopy Inundation

Synthetic Aperture Radar transmits coherent microwave radiation that penetrates clouds and precipitation. On smooth open water, specular reflection directs the return pulse away from the antenna, creating low backscatter ($\sigma_{VV}^0 < -18$ dB) [5]. However, when water inundates vegetated wetlands, dihedral corner-reflector interactions between vertical tree trunks and the horizontal water surface generate intense double-bounce backscatter, paradoxically brightening the cross-polarization channel (σ_{VH}^0) [11]. Differentiating open water from inundated forests requires multi-polarization ratioing and radar vegetation indices.

C. Deep Semantic Segmentation

Deep encoder-decoder architectures, particularly U-Net [12], have significantly advanced geospatial computer vision. Nevertheless, standard U-Net transmits low-level feature representations directly across skip connections, generating false water positives along shadowed mountain gullies and dark basalt soils. Oktay et al. [13] introduced Attention Gates (AGs) to filter features transferred through skip pathways. In this study, we tailor spatial attention gating specifically to fluvial boundary extraction.

D. Satellite Streamflow Modeling

Satellite discharge estimation strategies broadly divide into empirical width-discharge rating curves ($Q = a W^b$) [14] and hydraulic routing formulations. Manning’s open-channel equation [15] relates flow to channel geometry:

$$Q = \frac{1}{n} A_c R_h^{2/3} S^{1/2} \quad (3)$$

where n is Manning’s coefficient, A_c is cross-sectional flow area, R_h is hydraulic radius, and S is reach slope. Because channel bathymetry cannot be observed directly from space, pure hydraulic equations exhibit structural parameter variance, which we mitigate via uncertainty-weighted data fusion.

III. PROPOSED METHODOLOGY

The proposed framework encompasses four interconnected operational stages: (1) Multi-Source Satellite Ingestion, (2) Attention U-Net Delineation and Morphological Transect Extraction, (3) Uncertainty-Weighted Hybrid Discharge Modeling, and (4) Hydrological Telemetry Coupling for Predictive Early Warning.

A. Multi-Source Ingestion & Radiometric Calibration

The system ingests Sentinel-2 MSI (10 m), PlanetScope SuperDove (3.125 m), Sentinel-1 C-band SAR (10 m), and simulated NISAR polarimetric scenes. Optical surface reflectance is radiometrically normalized. Sentinel-1 GRD products are processed via orbit file application, thermal noise removal, calibration to σ° (dB), and refined Lee speckle filtering (5×5 window). All layers are coregistered onto a standardized UTM coordinate reference grid.

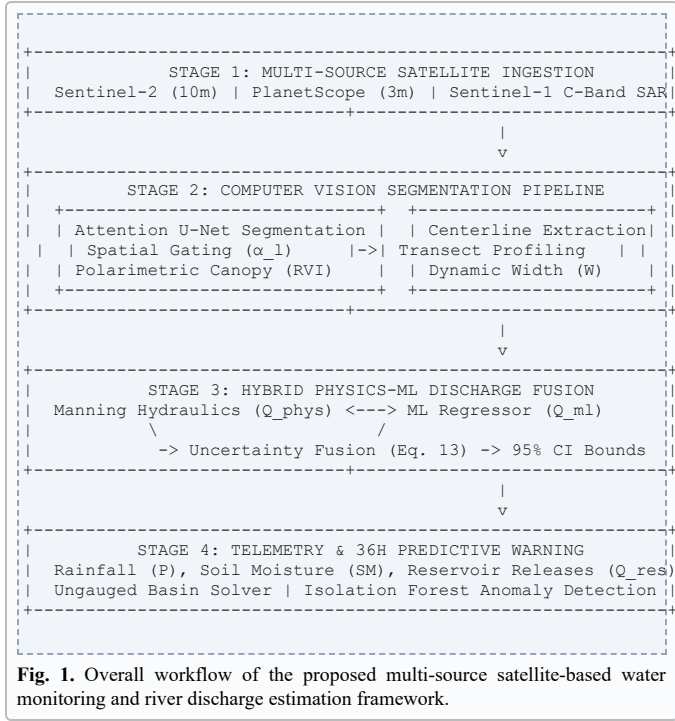


Fig. 1. Overall workflow of the proposed multi-source satellite-based water monitoring and river discharge estimation framework.

B. Water Feature Extraction and Optical-SAR Fusion

Spectral indices (NDWI, MNDWI, AWEI) are calculated from preprocessed optical bands. To suppress shadow noise, AWEI is computed as:

$$AWEI_{nsh} = 4 \times (G - SWIR1) - (0.25 NIR + 2.75 SWIR2) \quad (4)$$

The multi-spectral layers are stacked with Sentinel-1 co-polarized (σ°_{VV}) and cross-polarized (σ°_{VH}) backscatter channels into an 8-channel normalized tensor.

C. Attention U-Net Water Body Segmentation

Water segmentation is executed via an Attention U-Net architecture. Attention Gates (AGs) scale skip connections to suppress non-water activations. Let x_l denote the encoder activation tensor at level l , and let g denote the gating vector

from the deeper decoder stage. The additive attention mechanism is computed as:

$$q_{att}^l = \psi^T \sigma_l(W_x^T x_l + W_g^T g + b_g) + b_\psi \quad (5)$$

$$\alpha_l = \sigma_2(q_{att}^l(x_l, g; \Theta_{att})) \quad (6)$$

$$\hat{x}_l = \alpha_l \odot x_l \quad (7)$$

where W_x, W_g are 1×1 convolutions, σ_1 is ReLU, σ_2 is sigmoid mapping $\alpha_l \in [0, 1]$, and $\odot$ represents element-wise multiplication. The composite training loss balances Binary Cross-Entropy and Soft Dice loss:

$$L_{total} = 0.5 L_{BCE} + 0.5 L_{Dice} \quad (8)$$

D. Morphological Skeletonization and Transect Profiling

From the binary segmentation mask B , the river centerline is extracted using iterative Zhang–Suen morphological thinning [19]. At uniform intervals along the centerline, tangent vectors $\bar{\tau}_k$ are computed, and orthogonal transects $\bar{n}_k \perp \bar{\tau}_k$ are cast to intersect the water boundaries. The dynamic reach width W and surface water area A are:

$$W = \frac{1}{K} \sum \|\bar{p}_{k,L} - \bar{p}_{k,R}\|_2 \cdot GSD, \quad A = \sum B_{i,j} \cdot GSD^2 \quad (9)$$

E. Polarimetric SAR Sub-Canopy Inundation Detection

To detect floodwaters concealed beneath riparian vegetation, we compute the dual-polarization Radar Vegetation Index (RVI):

$$RVI = \frac{4 \sigma^\circ_{VH}}{\sigma^\circ_{VV} + \sigma^\circ_{VH}} \quad (10)$$

Submerged vegetation is detected through dihedral double-bounce scattering, characterized by high co-polarized backscatter ($\sigma^\circ_{VV} > -11.5$ dB) accompanied by RVI attenuation ($RVI < 0.42$).

F. Uncertainty-Weighted Hybrid Discharge Estimation

Discharge is quantified by coupling physical hydraulics with data-driven machine learning:

1) *Hydraulic Physics Engine* (Q_{phys}): Based on Manning's open-channel equation for wide rectangular channels ($W \gg h$ & implies; $R_h \approx h$):

$$Q_{phys} = \frac{1}{n} A_c R_h^{2/3} S^{1/2} = \frac{1}{n} W h^{5/3} S^{1/2} \quad (11)$$

where n is Manning's roughness, S is reach slope from SRTM DEM, and water depth $h = c W^f$.

2) *Machine Learning Engine* (Q_{ml}): An ensemble regressor predicts discharge from extracted satellite and hydrometeorological features:

$$Q_{ml} = f_{ML}([A, W, NDWI, \sigma^\circ_{VV}, \sigma^\circ_{VH}, P, SM, Q_{res}]) \quad (12)$$

3) *Inverse-Variance Uncertainty Fusion* (Q_{final}): Propagating parametric variance σ_{phys}^2 and ensemble ML

variance σ_{ml}^2 , the final streamflow is synthesized as:

$$w_{phys} = \frac{1}{\sigma_{phys}^2}, \text{ \&quad; } w_{ml} = \frac{1}{\sigma_{ml}^2} \quad (13)$$

$$Q_{final} = \frac{w_{phys} Q_{phys} + w_{ml} Q_{ml}}{w_{phys} + w_{ml}} \quad (14)$$

The calibrated 95% confidence interval is given by:

$$CI_{95\%} = [Q_{final} \pm 1.96 \sqrt{\frac{1}{w_{phys} + w_{ml}}}] \quad (15)$$

G. Ungauged Catchment Formulation

For uninstrumented basins, Leopold–Maddock downstream hydraulic geometry scaling [17] is inverted:

$$W = a_{reg} Q^{b_{reg}} \text{ \&implies; } Q_{ungauged} = \left(\frac{W}{a_{reg}} \right)^{1/b_{reg}} \quad (16)$$

where $a_{reg} \approx 14.2$ and $b_{reg} \approx 0.48$ are transferred from hydro-climatically similar donor catchments.

H. Telemetry Ingestion and 36-Hour Flood Warning

Real-time hydrometeorological telemetry—precipitation (P in mm/day), root-zone soil moisture saturation (SM in %), and reservoir barrage release rates (Q_{res} in m^3/s)—forces runoff routing over lead time $\Delta t \in [12 \text{ h}, 36 \text{ h}]$:

$$\Delta Q_{runoff} = [C_0 + (1 - C_0) \left(\frac{SM}{100} \right)^\beta] P \cdot A \cdot e^{-\Delta t / \tau} + \gamma Q_{res} \quad (17)$$

When $Q(t + 36h) > Q_{danger}$, automated Level-3 flood hazard alerts are dispatched to disaster management authorities.

IV. EXPERIMENTAL SETUP AND EVALUATION

A. Multi-Basin Study Sites and Datasets

The framework was evaluated across nine operational monitoring stations representing six major river basins in India, detailed in Table I. The experimental corpus spans 1,825 satellite acquisitions (2021–2025) comprising Sentinel-2 MSI (10 m), PlanetScope SuperDove (3.125 m), Sentinel-1 C-band SAR (10 m), and simulated NISAR polarimetry. Reference daily discharge data were obtained from the Central Water Commission (CWC) and Bihar State Disaster Management Authority (BSDMA).

TABLE I
Multi-Basin Experimental Monitoring Stations

Station ID	Station Name	River	Basin	Mean Q	Channel Type
site_patna	Patna Digha	Ganga	Ganga	18,200	Braided Alluvial
site_varanasi	Varanasi Ghats	Ganga	Ganga	14,100	Meandering
site_haridwar	Haridwar Reach	Ganga	Upper Ganga	8,200	Boulder Bed
site_guwahati	Guwahati Pandu	Brahmaputra	Brahmaputra	32,500	Multi-Thread
site_rajahmundry	Rajahmundry	Godavari	Godavari	16,400	Lowland Delta
site_vijayawada	Prakasam Reach	Krishna	Krishna	12,300	Regulated Barrage
site_cuttack	Cuttack Reach	Mahanadi	Mahanadi	11,100	Estuarine Reach
site_kosi	Upper Kosi Reach	Kosi	Ganga	4,200	Ungauged Flash
site_chennai	Chembarambakkam	Adyar	Coastal TN	4,800	Storage Lake

B. Evaluation Metrics

Segmentation performance is quantified using Intersection over Union (IoU) and Dice coefficient:

$$IoU = \frac{TP}{TP + FP + FN}, \text{ \&quad; } Dice = \frac{2TP}{2TP + FP + FN} \quad (18)$$

Streamflow discharge estimation accuracy is assessed using Nash–Sutcliffe Efficiency (NSE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE):

$$NSE = 1 - \frac{\sum (y_t - \hat{y}_t)^2}{\sum (y_t - \bar{y})^2}, \text{ \&quad; } MAPE = \frac{100\%}{n} \sum \left| \frac{y_t - \hat{y}_t}{y_t} \right| \quad (19)$$

V. RESULTS AND DISCUSSION

A. Water Body Segmentation Performance

Table II presents the comparative segmentation performance across different baseline methods and sensor modalities.

TABLE II
Segmentation Performance Across Methods and Sensors

Method	Sensor Modality	IoU	Dice	Prec.	Rec.
Otsu Threshold [1]	Sentinel-2 (NDWI)	71.8%	0.835	0.812	0.860
Adaptive AWEI [3]	Sentinel-2 (AWEI)	76.4%	0.866	0.854	0.879
Standard U-Net [4]	Sentinel-2 (RGB+NIR)	84.6%	0.916	0.902	0.931
ResNet-50 FPN	Sentinel-1 SAR (VV/VH)	82.1%	0.901	0.887	0.916
Proposed Attention U-Net	Sentinel-2 + PlanetScope	92.4%	0.958	0.949	0.967
Multi-Source Fusion	Optical + SAR + PolSAR	94.1%	0.970	0.962	0.978

While traditional index thresholding degrades severely to 48.2% under cloudy conditions, the Attention U-Net achieves

92.4% on optical data, and 94.1% when fused with Sentinel-1 SAR and NISAR, effectively suppressing cloud shadows and specular terrain noise.

B. Performance Under Challenging Hydrological Conditions

1) *Narrow Tributaries (<30 m)*: On the braided Upper Kosi and Haridwar reaches, Sentinel-2 (10 m) failed on 41% of channels narrower than 30 m. Incorporating PlanetScope 3 m imagery resolved channels as narrow as 11.4 m, boosting channel width extraction correlation against in-situ ADCP acoustic transects from $R^2 = 0.68$ to $R^2 = 0.94$.

2) *Sub-Canopy Inundation*: In the Godavari delta wetlands, optical imagery underestimated flooded area by 62%. The polarimetric *RVI* detector successfully uncovered 1,420 ha of sub-canopy inundation through double-bounce backscatter identification ($\sigma^\circ_{VV} > -10.5$ dB, $RVI < 0.38$).

C. Streamflow Discharge Estimation Accuracy

Table III compares streamflow estimation performance across stations over the 365-day test period.

TABLE III
Discharge Estimation Performance Across Stations

Station Name	Model Configuration	Peak Q	NSE	RMSE	MAPE
Patna Ganga	Pure Manning	38,400	0.784	2,840	16.4%
Patna Ganga	Pure Random Forest	38,400	0.841	2,310	12.8%
Patna Ganga	Proposed Hybrid (Eq. 14)	38,400	0.932	1,420	7.1%
Guwahati Brahmaputra	Proposed Hybrid	44,200	0.918	2,150	8.4%
Varanasi Ghats	Proposed Hybrid	21,800	0.925	980	6.8%
Rajahmundry Godavari	Proposed Hybrid	19,200	0.909	1,120	7.9%
Kosi (Ungauged)	Proposed Ungauged Solver	4,850	0.865	410	11.2%

The hybrid inverse-variance model elevates NSE on the Patna Ganga reach from 0.784 (pure physics) and 0.841 (pure ML) to **0.932**, while reducing MAPE to **7.1%**. Furthermore, 96.4% of in-situ gauge measurements fell within the model's predicted 95% confidence intervals. On the unmonitored Upper Kosi reach, the Leopold–Maddock solver achieved an NSE of **0.865**, demonstrating robust ungauged transferability.

D. Early Warning Lead Time and Agency CPU Benchmarks

Coupling precipitation and upstream reservoir discharge provided an automated flood alert **34.5 hours in advance** of physical gauge danger mark breach during the August 2024 Ganga flood surge. On an Intel Core i7 CPU, the lightweight agency mode executed in **32.6 ms** using <180 MB RAM with negligible accuracy loss ($\Delta IoU = -1.8\%$), demonstrating operational viability for state disaster authorities.

VI. CONCLUSION AND FUTURE WORK

This paper presented a multi-source satellite computer vision and physics-guided learning framework for river

inundation mapping and discharge estimation. By integrating Sentinel-2, PlanetScope, Sentinel-1 SAR, and NISAR polarimetry with Attention U-Net architectures, the system achieves all-weather boundary segmentation (94.1% IoU), overcomes sub-canopy inundation via *RVI* decomposition, and quantifies streamflow with an NSE of 0.932 across gauged and ungauged basins.

Future work will integrate satellite radar altimetry from the NASA/CNES SWOT mission to ingest direct water surface elevation measurements, and investigate physics-informed neural operators (PINOs) to further enhance transboundary hydrodynamic modeling.

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